

S. D. S. M. SCHOOL FOR EXCELLENCE

Practical questions of Python: STD XI (26-27)

Term1: Half yearly Program list

Rules for writing / Preparing Practical Note Copy

1. Practical Note book, one side Blank and one side rule
2. Each page must have page number, date and experiment number
3. In Experiment sub heading, write question (Experiment / Program)
4. Each program must have an output.
5. Output must be written on left side blank page in pencil.
6. Do not use Red, Green or Black pen.
7. Index page must be duly filled with page number and date.
8. Notebook must be covered by brown cover page.
9. In Certificate page, only write your name, std. and roll number.
10. All pages must have page number, experiment no and date

Write python program for the following questions:

Sequential Concept Based Program:

1. Write python code to display a welcome message in your desktop screen.
2. Accept your name and Admno and print as follows
 - a) Amit Kr 1234
 - b) Amit Kr
1234
 - c) Amit Kr
1234
3. Accept three integer numbers and print their SUM, and Multiplication values.
4. Enter your Name, Age and Address with Phone Number. Print them.
5. Accept your name, Std and section with Marks of English, Physics, Chemistry, Mathematics and Computer Science. Display your name, Std and section with Total marks obtained.
6. Enter Cost Price and Sell Price and find the Profit Amount and profit Percentage.
7. Enter Principle, rate and time and calculate Simple Interest Amount.
8. Enter Principle, rate and time and calculate Compound Interest Amount.
Formula of CI is $P (1 + r/100)^t - P$
9. Calculate the root of quadratic equation $r1 = -b + \sqrt{b^2 - 2ac} / 2a$ and $r2 = -b - \sqrt{b^2 - 2ac} / 2a$
10. Write python program to solve these equations
$$A = \frac{v - u}{t}$$
11. Calculate Area of Circle, Area of Square and Perimeter of Rectangle.
12. Convert : Hour into Minutes, KG into gram, Rupee into Paisa

Conditional Statement

13. Enter a number and check it is even or odd number.
14. Enter a number and check it is divisible by 7 or not.
15. Enter a number and check it is divisible by 5 and 7 both, divisible by 5 only, 7 only or none.
16. Enter two integer numbers and print the highest one.
17. Enter three integer values, print the highest value, and smallest value.
18. Enter Sell price and Cost price and find profit or loss percentage.
19. Write a menu driven program, on the basis of your selection, perform Addition, Subtraction, Multiplication and Division program.
20. Accept a letter and check it is in upper case, lower case.
21. Accept a character and check it is a letter, digit, space or special character.
22. Accept year and check it is leap year or not.
23. Enter your name, age & gender and check you are eligible to marry or not.
24. Write the program to find the ticket fair of a bus according to the code of a bus:

Code	Fair (in Rs)
D	10
R	15
W	20
c	25

25. On the basis of input value of avgmk, print grade of a student, if

Avgmk	Grade
<=40	'F'
40 to 50	E
50 to 60	D
60 to 75	C
75 to 85	B
85 to 95	A
95 to 100	A+

26. Calculate Electric bill for given meter reading for given details

Mtr Regd(in Unit)	Rupee per / Unit
Up to First 100 units	Rs. 120
Next 100 units	Rs. 1 per unit
Next 100 units	Rs. 1.25 per unit
Next 200 units	Rs. 1.50 per unit
Rest	Rs. 2 per unit

Accept Meter reading in Unit and Print the Bill Amount with 8% GST on Total Amount

Looping or Iteration Program

27. Print 1,2,3,4,5,6,7,8,9,10 using for loop
28. Print 10,9,8,7,6,5,4,3,2,1
29. Print series 2,4,6,8,10....50
30. Print series 1,4,7,10,13,16,19
31. Print series 1,4,9,16,25...100
32. Print series 1,10,2,9,3,8,4,7....10,1
33. Print multiple table of 5.
34. Print multiple table of n.
35. Print all odd numbers between 1 and n.
36. Print 1,2,3,4,5,6,7,8,9,10 using while loop
37. Print Series 1,2,3,5,6,7,9,10 using jump statement
38. Print Fibonacci series up to 50 using while loop.
39. Print the sum of all even number present between 10 to 50
40. Enter 10 different numbers and print the sum of all odd number and their average using while loop.

41. Write a program to input the value of x and n and print the sum of the given series

$$S=1-x+x^2-x^3+x^4+ \dots x^n$$

42. Write a program to input the value of x and n and print the sum of the given series

$$S=x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \frac{x^n}{n!}$$

43. Print the sum of all numbers from n1 to n2 where n2>n1

44. *	AAAAA	1	1	1234
**	AAA	12	23	234
***	A	123	456	34
****		1234	7890	4

45. Check the entered number is Prime, Palindrome or none.
46. Enter a number and print its reverse, e.g., 235, output will be 532 using while loop.
47. Enter a number of three digits only, and print the sum of its digits.
48. Enter a decimal number and print its octal value.
49. Enter a string and check it is Palindrome or not.
eg MADAM is a palindrome word as its reverse is same as original
50. Enter a number and check it is Twisted Prime or Not.
(Note: A number is said to be twisted prime if it is a **prime number** and reverse of the number is also a prime number, eg 13, 79)
51. Enter a number and check it is perfect number or not.
(Note: perfect number, a positive integer that is equal to the sum of its proper divisors.)

52. Enter a sentence and print its words in different lines
53. Enter a sentence and count number of words in it.
54. Count and display the number of vowels, consonants, uppercase, lowercase characters in given string
55. Enter an integer number and print all its factors
56. Enter a number and find whether the entered number is Prime or not.
57. Print all prime numbers between 10 to 100.
58. Write a program to whether a number is an Armstrong number or not.
A number is said to be an Armstrong, if the sum of the cubes of the digit of the number is the same as the original number. Eg $153 = 1^3 + 5^3 + 3^3$
59. Check whether the entered number is Duck number or not.
A **duck number** is a positive number that contains at least one zero digit, provided the zero does not appear at the very beginning of the number. Duck numbers are 570, 7067, and 890 and 567 are not duck numbers
60. Print multiple tables of numbers from 1 to 10